The Convex Origin of Fixed Costs[1]
Online Appendix

Contents

[1]Hanbaek Lee, University of Cambridge, Email: hl610@cam.ac.uk

A Proof of Proposition 4 (Multi-action duality)

Proof.
Part (i): General case. The agent observes cost draws $(\xi_1, \ldots, \xi_J)$ with $\xi_j \sim G_j$ independently, and selects the action with the highest net payoff:

$$\Pi^{FC}(\mathbf{S}) = E\left[\max_{j\in\{0,\ldots,J\}} \{S_j - w\xi_j\}\right].$$

Partition the expectation by which action is optimal. For each $j \geq 1$, action j is chosen when $S_j - w\xi_j \geq S_k - w\xi_k$ for all $k \neq j$ and $S_j - w\xi_j \geq 0$. Under independence, the probability that action j is chosen can be written as a function $\varphi_j(\mathbf{S})$, and the expected cost paid conditional on choosing j is $E[w\xi_j \mid j \text{ chosen}]$.

Define $\varphi_0 = 1 - \sum_{j=1}^{J} \varphi_j$ (the probability of inaction). The expected payoff is:

$$\Pi^{FC}(\mathbf{S}) = \sum_{j=0}^{J} \varphi_j^* S_j - \sum_{j=1}^{J} E[w\xi_j \cdot \mathbf{1}_{j \text{ chosen}}].$$

By the Fenchel conjugate construction, $\Pi^{FC}(\mathbf{S}) = \sup_{\boldsymbol{\varphi}\in\Delta^J}\{\sum_j \varphi_j S_j - C(\boldsymbol{\varphi})\}$ where $C(\boldsymbol{\varphi})$ is the convex conjugate of Π^{FC}. Under independence of the G_j, the cost function inherits a separable structure: the marginal cost of raising φ_j depends only on G_j, not on the other distributions.

Joint convexity of C follows from the fact that $\Pi^{FC}(\mathbf{S})$, as a pointwise maximum of affine functions of $\mathbf{S}$, is convex in $\mathbf{S}$. The Fenchel biconjugate theorem then implies that $C = (\Pi^{FC})^*$ is convex on Δ^J.

Part (ii): i.i.d. Gumbel specialisation. Let $\xi_j \overset{i.i.d.}{\sim} \text{Gumbel}(\mu, \sigma)$ for $j = 0, 1, \ldots, J$, with CDF $G(\xi) = e^{-e^{-(\xi-\mu)/\sigma}}$. The expected maximum of $\{S_j - w\xi_j\}$ is a classical result (**?**):

$$\Pi^{FC}(\mathbf{S}) = \sigma \ln\left(\sum_{j=0}^{J} e^{S_j/\sigma}\right) + \mu',$$

where $\mu' = w\mu - w\sigma\gamma_E$ and γ_E is the Euler–Mascheroni constant. This is the log-sum-exp function (up to constants), which is convex in $\mathbf{S}$.

The gradient gives the choice probabilities:

$$\varphi_j^* = \frac{\partial \Pi^{FC}}{\partial S_j} = \frac{e^{S_j/\sigma}}{\sum_{k=0}^{J} e^{S_k/\sigma}},$$

the multinomial logit. The Fenchel conjugate of the log-sum-exp is the negative

entropy:

$$C(\boldsymbol{\varphi}) = \sup_{\mathbf{s}} \left\{ \sum_j \varphi_j S_j - \sigma \ln\left(\sum_j e^{S_j/\sigma} \right) \right\} - \mu' = \sigma \sum_{j=0}^{J} \varphi_j \ln \varphi_j,$$

which is non-negative on Δ^J (since $\ln \varphi_j \leq 0$) and jointly convex. ■

Remark 1. *The Gumbel distribution has unbounded support $(-\infty, \infty)$, so negative cost draws are possible. In the original fixed cost interpretation, this means that some realisations yield a "bonus" rather than a cost. The duality holds provided $E[\max_j\{S_j - w\xi_j\}]$ is finite, which is guaranteed by the double-exponential tails of the Gumbel. For distributions on $[0, \bar{\xi}]$ (bounded support), the convergence is automatic.*

B Proof of Proposition 10 (Nonlinear Phillips curve)

Proof.
Assume that f_{ss} is continuously differentiable with $\int |x|\, f_{ss}(x)\, dx < \infty$ and that G is C^1 with bounded density g, so that differentiation under the integral sign is justified by the dominated convergence theorem. Write $f_t(x) = f_{ss}(x - \mu_t)$. Then

$$\pi_t = \int G\left(\frac{Bx^2}{w}\right) x \cdot f_{ss}(x - \mu_t)\, dx.$$

Differentiating with respect to μ_t using Leibniz's rule:

$$\frac{\partial \pi_t}{\partial \mu_t} = -\int G\left(\frac{Bx^2}{w}\right) x \cdot f'_{ss}(x - \mu_t)\, dx,$$

since $\frac{\partial}{\partial \mu_t} f_{ss}(x - \mu_t) = -f'_{ss}(x - \mu_t)$. Evaluating at $\mu_t = 0$:

$$\left.\frac{\partial \pi_t}{\partial \mu_t}\right|_{\mu_t = 0} = -\int G\left(\frac{Bx^2}{w}\right) x \cdot f'_{ss}(x)\, dx.$$

Now integrate by parts with $u = G(Bx^2/w) \cdot x$ and $dv = f'_{ss}(x)\, dx$:

$$-\int u\, dv = -\left[G\left(\tfrac{Bx^2}{w}\right) x \cdot f_{ss}(x)\right]_{-\infty}^{\infty} + \int f_{ss}(x) \frac{d}{dx}\left[G\left(\tfrac{Bx^2}{w}\right) x\right] dx.$$

The boundary term vanishes because f_{ss} has tails decaying faster than any polynomial. Expanding the derivative:

$$\frac{d}{dx}\left[G\left(\frac{Bx^2}{w}\right)x\right] = g\left(\frac{Bx^2}{w}\right)\frac{2Bx}{w}\cdot x + G\left(\frac{Bx^2}{w}\right) = g\left(\frac{Bx^2}{w}\right)\frac{2Bx^2}{w} + G\left(\frac{Bx^2}{w}\right).$$

Substituting:

$$\kappa(G) \;=\; \left.\frac{\partial \pi_t}{\partial \mu_t}\right|_{\mu_t=0} \;=\; \int G\left(\frac{Bx^2}{w}\right) f_{ss}(x)\,dx + \int g\left(\frac{Bx^2}{w}\right)\frac{2Bx^2}{w}\, f_{ss}(x)\,dx.$$

The first integral is the extensive margin: holding the hazard fixed, shifting the gap distribution by μ_t shifts the average price change by the average hazard. The second integral is the intensive margin: the shift changes the surpluses of firms near the threshold, changing which firms adjust. Since $G(Bx^2/w)$ is even in x and f_{ss} is symmetric around zero, the integrand $G(Bx^2/w)\cdot x\cdot f_{ss}(x)$ is odd and integrates to zero in steady state, confirming that $\pi = 0$ at $\mu_t = 0$; the linearisation coefficient κ is well-defined and given by the formula above. ■

C Proof of Corollary 5 ($\kappa = (1+2\gamma)\bar{\Lambda}$ for power laws)

Proof.
For $G(\xi) = (\xi/\bar{\xi})^\gamma$, the density is $g(\xi) = (\gamma/\bar{\xi})(\xi/\bar{\xi})^{\gamma-1}$. Evaluate the intensive margin integrand at $\xi = Bx^2/w$:

$$\begin{aligned} g\left(\frac{Bx^2}{w}\right)\cdot\frac{2Bx^2}{w} &= \frac{\gamma}{\bar{\xi}}\left(\frac{Bx^2/w}{\bar{\xi}}\right)^{\gamma-1}\cdot\frac{2Bx^2}{w} \\ &= 2\gamma\cdot\frac{1}{\bar{\xi}}\left(\frac{Bx^2/w}{\bar{\xi}}\right)^{\gamma-1}\cdot\frac{Bx^2}{w} \\ &= 2\gamma\left(\frac{Bx^2}{w\bar{\xi}}\right)^{\gamma} = 2\gamma\,\Lambda(x). \end{aligned} \tag{1}$$

Substituting into equation (20) of the main paper:

$$\begin{aligned} \kappa(G) &= \int G\left(\frac{Bx^2}{w}\right) f_{ss}(x)\,dx + \int 2\gamma\,\Lambda(x)\, f_{ss}(x)\,dx \\ &= \bar{\Lambda} + 2\gamma\bar{\Lambda} = (1+2\gamma)\bar{\Lambda}. \end{aligned}$$

The selection wedge is $\kappa - \bar{\Lambda} = 2\gamma\bar{\Lambda}$. The ratio $\kappa/\bar{\Lambda} = 1 + 2\gamma$ depends only on γ, not on f_{ss}, B, or w. ■

D Proof of Corollary 4 (Ordering of non-neutrality)

Proof.
The real effect of a monetary shock dm is proportional to $(1-\kappa)\,dm$, so non-neutrality is decreasing in κ.

(i) Calvo benchmark. When Λ is constant at λ, the intensive margin vanishes: $g(Bx^2/w) \cdot 2Bx^2/w = 0$ for all x in the Calvo limit, since a constant hazard implies $g = 0$ almost everywhere. Therefore $\kappa = \bar{\Lambda} = \lambda$ and the selection wedge is zero. Non-neutrality is $1 - \lambda$, the maximum among models with the same frequency λ.

(ii) Absolutely continuous G with positive density. By Proposition 1(iii), $g > 0$ implies $C'' > 0$, so $\Lambda(x)$ is strictly increasing in $|x|$. The intensive margin $\int g(Bx^2/w)(2Bx^2/w)f_{ss}(x)\,dx > 0$ since the integrand is non-negative and strictly positive on a set of positive f_{ss}-measure. Therefore $\kappa > \bar{\Lambda}$ and the selection wedge is strictly positive.

(iii) Golosov-Lucas. When G is degenerate at $\bar{\xi}$, $\Lambda(x) = \mathbf{1}[Bx^2 \geq w\bar{\xi}]$. All adjusters have gaps at or beyond the threshold $\bar{x} = \sqrt{w\bar{\xi}/B}$, so the intensive margin is maximised (concentrated entirely at the boundary). Non-neutrality is minimal.

Monotonicity in γ: For the power law family, $\kappa = (1+2\gamma)\bar{\Lambda}$ by Corollary 5. At fixed $\bar{\Lambda}$, κ is strictly increasing in γ and non-neutrality $1-\kappa$ is strictly decreasing. ■

E Proof of Proposition 9 (Selection and the menu cost distribution)

Proof.
Write the adjustment-weighted mean absolute gap as

$$\overline{|x|}^{adj} = \frac{E[\Lambda(x) \cdot |x|]}{E[\Lambda(x)]} = E[|x|] + \frac{\mathrm{Cov}(\Lambda(x)/\bar{\Lambda},\ |x|)}{1}.$$

The covariance is positive when Λ is increasing in $|x|$. Its magnitude depends on how sharply Λ responds to $|x|$.

(i) The elasticity of the hazard with respect to $|x|$ is

$$\frac{\partial \ln \Lambda(x)}{\partial \ln |x|} = \frac{g(Bx^2/w)}{G(Bx^2/w)} \cdot \frac{2Bx^2}{w},$$

which is twice the hazard rate of G evaluated at the firm's surplus. A higher hazard rate means Λ responds more sharply to $|x|$, increasing the covariance between $\Lambda(x)/\bar{\Lambda}$ and $|x|$, and hence increasing $\mathcal{S}$.

(ii) By Proposition 1(iii), $C'' = w/g(G^{-1}(\varphi))$. When g is small at the threshold, C'' is large and Λ responds steeply to changes in x^2: few agents are clustered near the margin, so a small increase in the surplus tips few new firms into adjustment but those who do adjust have substantially larger gaps. This concentrates adjustment among high-gap firms, strengthening selection.

(iii) For constant Λ (Calvo), the covariance is zero and $\mathcal{S} = 0$. For a step function Λ (Golosov-Lucas), adjustment is deterministic above the threshold: $\mathrm{Cov}(\Lambda/\bar{\Lambda}, |x|)$ is maximal since $\Lambda/\bar{\Lambda}$ perfectly separates high-gap from low-gap firms.

For the power law family, the hazard rate of G at ξ is $g(\xi)/G(\xi) = \gamma/\xi$, so the elasticity is 2γ — constant and strictly increasing in γ. ■

F Proof of Proposition 6 (Investment response and the cost density)

Proof.
By Theorem 1, the optimal investment probability is $\varphi^*(k) = G(S(k)/w)$. Differentiating with respect to the surplus:

$$\frac{d\varphi^*}{dS} = \frac{g(S/w)}{w} = \frac{g(G^{-1}(\varphi^*))}{w} = \frac{1}{C''(\varphi^*)},$$

where the last equality uses $C''(\varphi) = w/g(G^{-1}(\varphi))$ from Proposition 1(iii). This is the density of cost draws at the current threshold, scaled by $1/w$: the number of firms on the margin of investing per unit surplus.

Taking expectations across the cross-section of firms:

$$\frac{dE[\varphi^*(k)]}{dS} = E\left[\frac{g(G^{-1}(\varphi^*(k)))}{w}\right],$$

which is the expected density at the threshold, averaged over the distribution of capital stocks k. This is positive and proportional to the density of agents at the adjustment margin. ■

G Proof of Proposition 8 (Stationary distribution)

Proof.
The price gap evolves according to:

$$x_{t+1} = \begin{cases} z_{t+1} & \text{with probability } \Lambda(x_t), \\ x_t + z_{t+1} & \text{with probability } 1 - \Lambda(x_t), \end{cases}$$

where $z_{t+1} \sim \phi_z$ independently. The density of x_{t+1} conditional on the current cross-section f_t is:

$$f_{t+1}(x) = \int \Lambda(\tilde{x}) f_t(\tilde{x})\, d\tilde{x} \cdot \phi_z(x) + \int (1 - \Lambda(\tilde{x}))\phi_z(x - \tilde{x}) f_t(\tilde{x})\, d\tilde{x}.$$

The first term: adjusting firms (total mass $\bar{\Lambda}_t$) draw $z_{t+1} \sim \phi_z$ as their new gap. The second term: non-adjusting firms at $\tilde{x}$ receive shock z_{t+1} and move to $\tilde{x} + z_{t+1}$; the density at x is the convolution of $(1 - \Lambda(\tilde{x}))f_t(\tilde{x})$ with ϕ_z.

Setting $f_{t+1} = f_t = f_{ss}$ yields the fixed-point equation (13) of the main paper. Symmetry: if ϕ_z is symmetric around zero and $\Lambda(x) = \Lambda(-x)$ (which holds since $\Lambda(x) = G(Bx^2/w)$), then if f_t is symmetric, so is f_{t+1}. Since f_{ss} is the unique fixed point of this contraction (the operator is a weighted average of a density and a convolution, both preserving the space of symmetric densities), f_{ss} is symmetric around zero. ■

H Comparison with Williams-Daly-Zachary

Section 2.2 of the main paper discusses the relationship between Theorem 1 and the Williams-Daly-Zachary (WDZ) theorem (**??**). This appendix provides a systematic comparison. Both results share a common mathematical skeleton — an expected-maximum operator equals the Fenchel conjugate of a function determined by a distributional primitive — but differ in their economic content, scope, and implications. Table **??** summarises the comparison along six dimensions.

For binary choice ($J = 2$), the two results are formally equivalent under a sign change: a random benefit $S - w\xi$ versus a random cost $w\xi$ applied to a fixed benefit S are isomorphic. The binary conjugate identity is therefore not new. The contributions of this paper lie beyond the duality identity itself: the mapping $G \mapsto C$ and the distributional taxonomy (Propositions 3–5), the structural explanation for the **?** result, and the continuous parametrisation of the pricing spectrum are entirely absent from WDZ and the discrete choice literature.

Table 1: Williams-Daly-Zachary vs. Theorem 1

Dimension	**WDZ**	**Theorem 1**
Random element	Taste shocks ε_j on payoffs	Cost draws $w\xi$ on adjustment
Decision	Multi-alternative discrete choice	Binary: adjust or not
Surplus function	$E[\max_j\{u_j + \varepsilon_j\}]$	$E[\max\{S - w\xi,\ 0\}]$
Convexity result	Surplus convex in $(u_1, \ldots, u_J)$	Payoff convex in S
Conjugate of	Generating function $\mathcal{G}(\varepsilon)$	$C(\varphi) = \int_0^\varphi G^{-1}(t)\,dt$
Choice probs	$p_j = \partial E[\max]/\partial u_j$	$\varphi^* = G(S)$
Primitive	Joint distribution of ε	Distribution G of costs
Coupling	Not constructed	Path-by-path equivalence
Economic context	Demand: consumer choice	Supply: adjustment frictions
Application	Multinomial logit (**?**)	Calvo–GL spectrum

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